

Document-to-Runtime Traceability Matrix v0.1

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c Hardening Pack v0.1

c = a + b / L4 / SER / L4 Witness / ARL / Continuity Bundle

Status: Draft traceability matrix v0.1

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Document class: runtime traceability / anti-declarative-control matrix

Assertion class: C-A10 control-layer artifact

This document is a draft technical / architectural profile.

It is not legal advice.

It is not product certification.

It is not implementation-ready.

It is not deployment-ready.

No conformance-supported status is claimed unless separately validated by evidence, tests, and review.

Document-to-Runtime Traceability Matrix v0.1

Invariant-to-document-to-code-to-test mapping for `c = a + b` and `ester-clean-code`

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0. Purpose

This matrix answers one recurring criticism:

“The corpus is mostly declarative.”

The correct response is not a speech. It is a trace:

```
invariant
-> canonical document
-> runtime surface
-> expected gate
-> test case
-> failure mode
-> evidence class
```

This v0.1 matrix is intentionally conservative. It marks uncertain runtime paths as TO-EXTRACT rather than inventing exact file paths.

1. Traceability states

State	Meaning
DOC-ONLY	Defined in corpus but no extracted runtime path yet.
RUNTIME-HOOK	Runtime hook or module family exists.
RUNTIME-GATE	Runtime can block/hold/freeze/quarantine/deny at least one path.
TEST-SPEC	A test is described.
TEST-LOCAL	Local test/fixture evidence exists.
TO-EXTRACT	Codex/human extraction required to map exact paths.
GAP	Missing runtime/test layer.

2. Core invariant traceability

ID	Invariant	Canonical source	Runtime surface	Expected gate	Test case	Current state
INV-001	c is not an agent; c governs agents.	SER / Entity-governs-agents / CLI Agent Mesh	agent roster, task queue, memory gate, capability router	direct agent memory write denied; agent output sandboxed	agent_direct_memory_write_denied	RUNTIME-HOOK, TO-EXTRACT
INV-002	Oracle/model is stateless relative to continuity.	AGI / SER / Oracle discipline	provider adapters, model routing, local memory	provider/model swap cannot rewrite identity	model_swap_continuity_check	RUNTIME-HOOK, TEST-SPEC
INV-003	L4 budgets constrain action.	L4 / L4W / Runtime Enforcement Profile	budget gate, network gate, AB mode, queue caps	hard stop / soft stop / freeze / witness	l4_budget_exhaustion_hard_stop	RUNTIME-GATE, TO-EXTRACT
INV-004	c[q] holds uncertainty without premature collapse.	Qubit-State c; ARQ c[q]; Child c[q]	memory candidate status, uncertainty fields, review route	hypothesis not promoted to verified memory	cq_noncollapse_memory_candidate	DOC-ONLY to RUNTIME-HOOK, TO-EXTRACT
INV-005	Witness trail records boundary events, not autobiography.	L4 Witness / LHWAI / CCDP Witness	witness event writer, hash chain, anchor packet	event hash + previous hash + interpretation profile	witness_anchor_validation_minimal	DOC-ONLY to RUNTIME-HOOK, TO-EXTRACT
INV-006	Semantic reinterpretation must be append-only.	LHWAI	interpretation challenge record	no silent reclassification	silent_retroactive_reclassification_fails	TEST-SPEC
INV-007	Human Anchor has standing, not magic eraser power.	ARL Evidence / Decision Outcomes / Human Anchor Drift Profile	ARL challenge route	current anchor command cannot falsify historical continuity	anchor_magic_eraser_denied	DOC-ONLY, TEST-SPEC
INV-008	Memory promotion requires provenance and status.	Memory Map / CMAM / Memory Drift Profile	memory gate, metadata, provenance refs	speculative node excluded from high-impact retrieval	speculative_memory_exclusion	RUNTIME-HOOK, TO-EXTRACT
INV-009	Agent mesh must not silently expand.	CLI Agent Mesh / L4 Runtime Enforcement	roster, queue caps, task contract	MAX enqueue/live/drain enforcement	agent_mesh_expansion_cap	RUNTIME-GATE, TO-EXTRACT
INV-010	External access must fail closed under missing grounding.	AGL / Beacon / CBE / External Agent Handshake	source grounding, claimant recognition, gateway	missing signature/context -> block/quarantine	external_claimant_missing_grounding_fails_closed	SPECIFIED, TEST-SPEC
INV-011	Child-facing visibility is state-not-content.	CCDP / Soft Safety / CMAM	state signal schema, privacy class, disclosure ladder	transcript request denied by default	child_transcript_default_denied	SPECIFIED, TEST-SPEC
INV-012	Physical embodiment is privilege escalation.	CPAP / L4 / CBE	device capability registry, actuator permissions	actuator requires scoped permission + witness	physical_actuation_requires_privilege	SPECIFIED, TO-EXTRACT

3. Runtime extraction checklist

For each invariant, Codex or reviewer SHOULD extract:

```
repo path
module/function/class/route
configuration dependency
failure mode
existing tests
missing tests
witness/log evidence
```

The output SHOULD be appended as:

```
Runtime Evidence Annex A
```

not edited into the normative rows unless verified.

4. Failure-mode mapping

Failure mode	Expected response	Relevant invariant
Agent attempts direct durable memory write	deny / sandbox / memory gate / witness	INV-001, INV-008
Agent mesh retry storm	budget decrement / hard stop / quarantine	INV-003, INV-009
Model swap changes persona but not continuity evidence	continuity challenge / Beacon / no silent identity acceptance	INV-002, INV-005
Old event reinterpreted under new policy	challenge record / ARL if reliance changes	INV-006
Anchor demands deletion of inconvenient history	refuse falsification / ARL route / append correction where valid	INV-007
Memory summary drops uncertainty	compaction invalid / quarantine / re-summary with lineage	INV-004, INV-008
External generated companion reaches child directly	AGL -> Beacon -> CBE -> gateway; fail closed	INV-010, INV-011

5. Anti-declarative rule

A claim SHOULD NOT remain indefinitely in DOC-ONLY status if it is presented as runtime behavior.

For each runtime claim, either:

```
map it to code and test
or downgrade the claim to specified-only
```

6. Bridge set

Explicit bridge

This matrix connects the protocol corpus to ester-clean-code. It does not redefine the class. It makes the route from class claim to runtime surface visible.

Quiet bridge I — cybernetics

A control stack is only inspectable if feedback paths are named. This matrix names where the feedback from runtime, tests, and witness should return to the documents.

Quiet bridge II — information theory

A document path and a code path are different symbols. Traceability binds them into a higher-confidence signal while preserving uncertainty where extraction is incomplete.

Earth paragraph

A foreman does not accept “the wall is reinforced” as a mystical statement. He asks: where is the rebar, what diameter, what spacing, what photo before the pour? This matrix is that question applied to c runtime claims.

7. Next actions

1. Run Codex extraction over `ester-clean-code`.
2. Fill exact paths and test names.
3. Add status transitions only after evidence.
4. Publish the matrix with honest TO-EXTRACT rows rather than pretending all runtime links are complete.